

Introduction to Machine Learning and Artificial Intelligence

Machine Learning Methods

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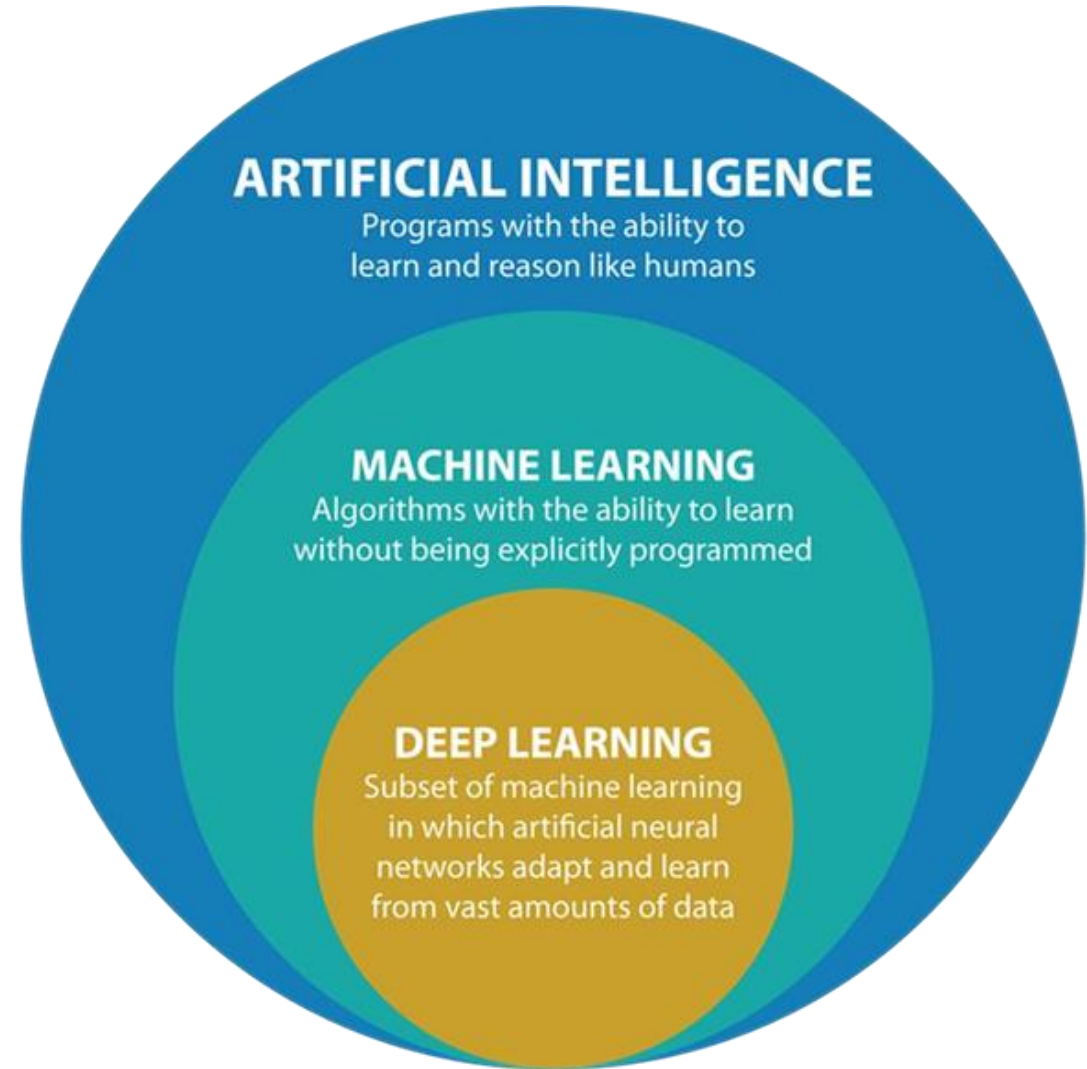
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Overview

- **Introduction to Machine Learning (ML)**
- **Types of ML**
- **How Machines Learn: Models and methods**
- **How to evaluate models**
- **Real world applications**

What is Machine Learning?

- Machine Learning was the term first introduced by Arthur Samuel in 1959.
- A subset of AI focused on systems that learn from data.
- Instead of being programmed explicitly, ML models find patterns and make predictions.
- The goal: Learn from data to improve automatically.
- Combines statistics, mathematics, and optimization with computing power.

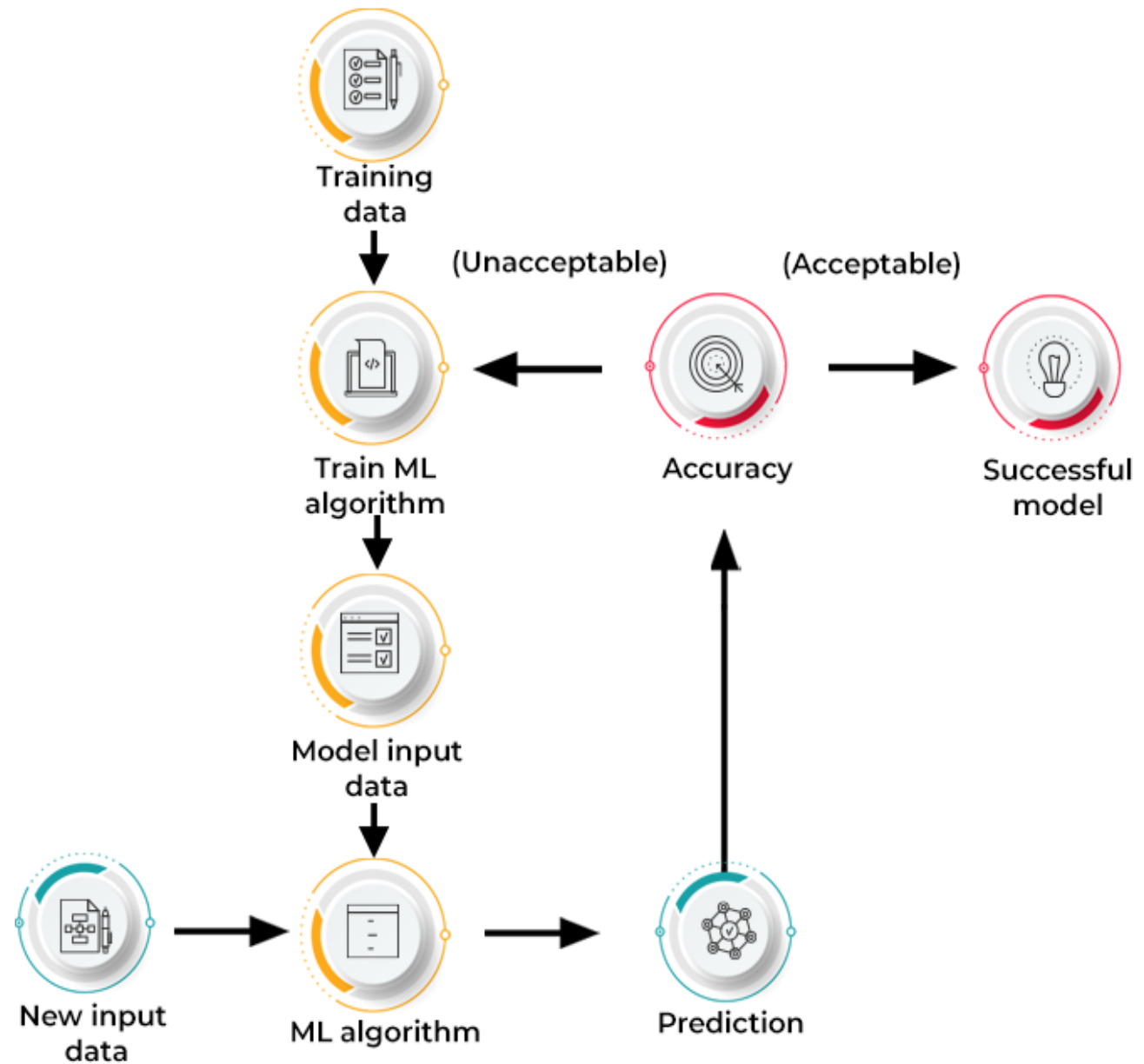


Source: www.argility.com

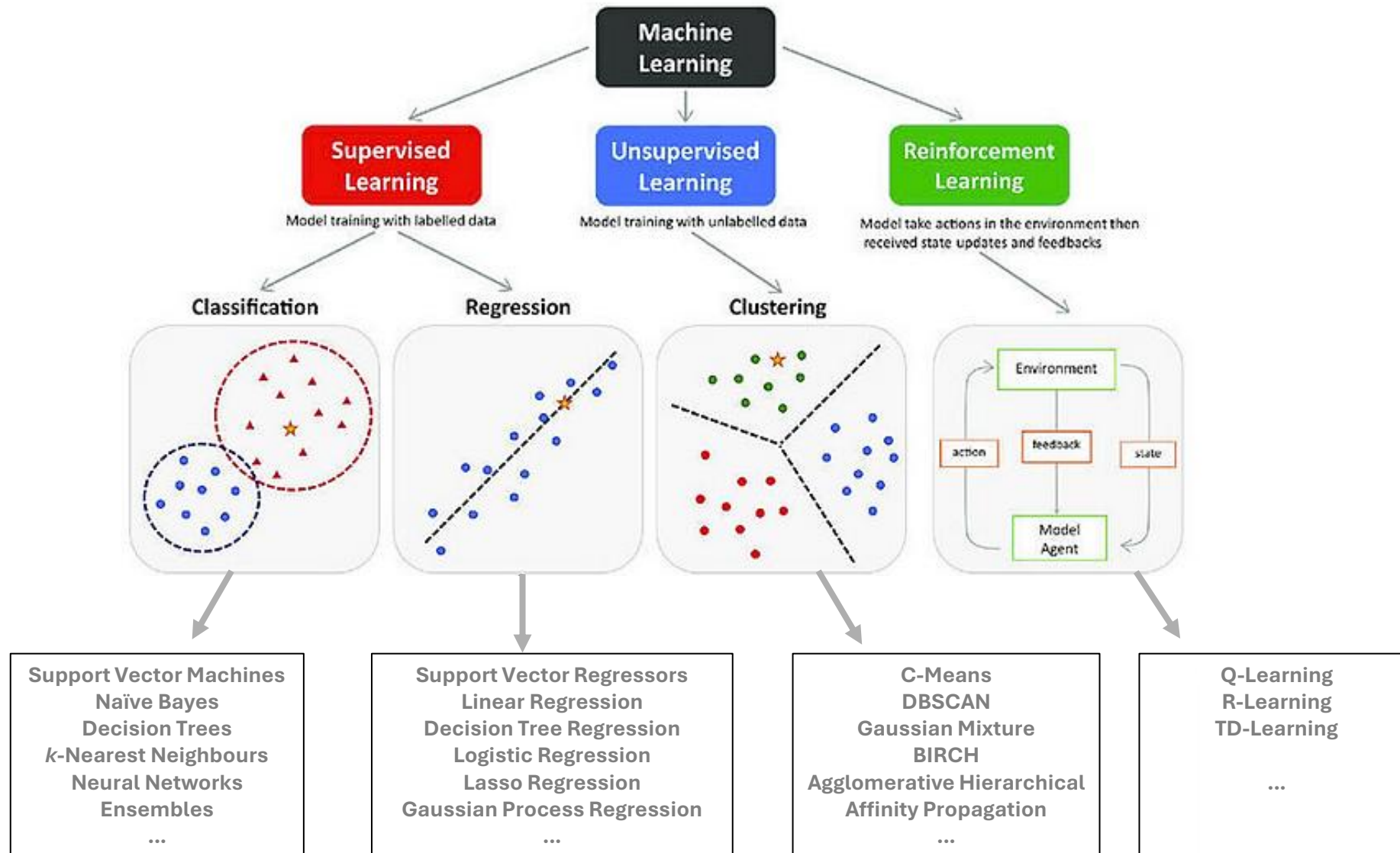
What is Machine Learning?

Machine Learning is a technology that enables computers to learn from given data and make predictions or decisions without being explicitly programmed.

The predictions or decisions involves training the machine algorithms on large datasets to recognize patterns and improve over time.

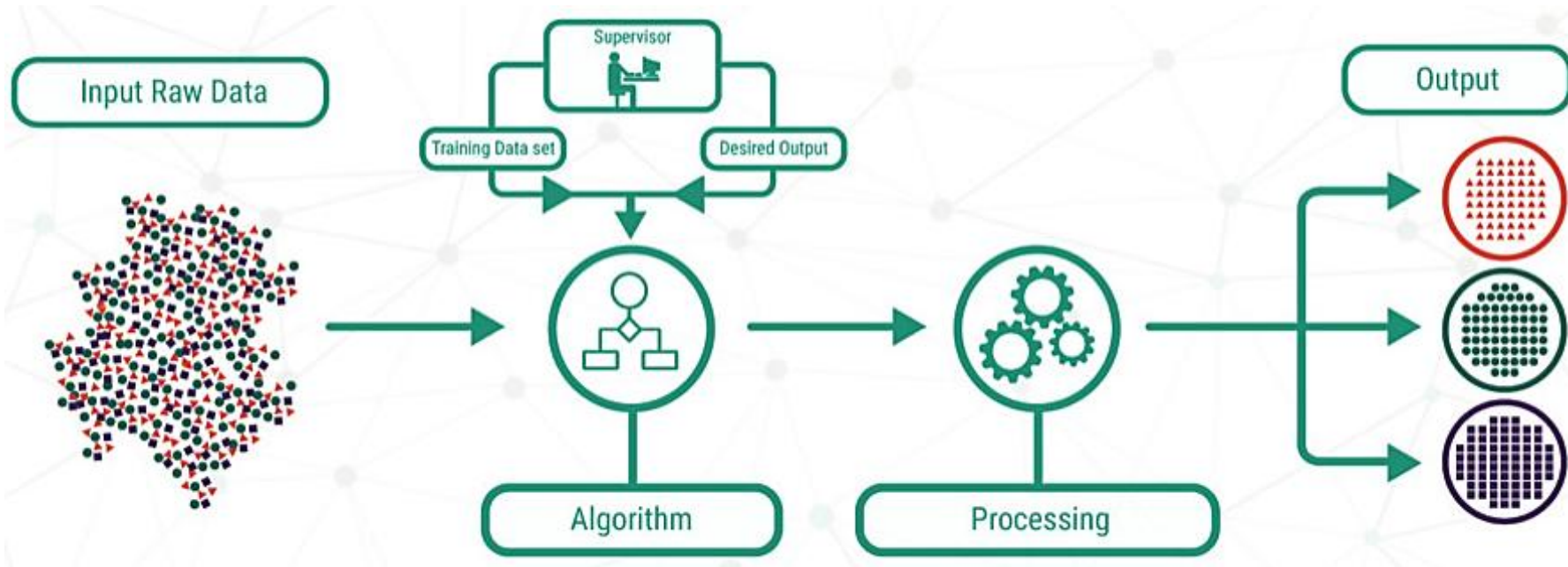


Types of Machine Learning



Supervised Learning

- Model is trained using labelled data.
- Each data point has a corresponding correct output, allowing the model to learn from past examples and make future predictions.
- The goal is to make accurate predictions or classifications on new, unseen data based on the patterns learned during training.



Source: [van Loon, R. \(2024\)](#)

Supervised Learning

Data Requirement: Requires a large amount of labelled data for training.

Model Types: **Classification** (categorical outputs) and **Regression** (continuous outputs).

Learning Process: The model is provided with both input data and corresponding output labels during training.

Evaluation: The performance is evaluated using metrics such as accuracy, precision and mean squared error (MSE).

Applications:

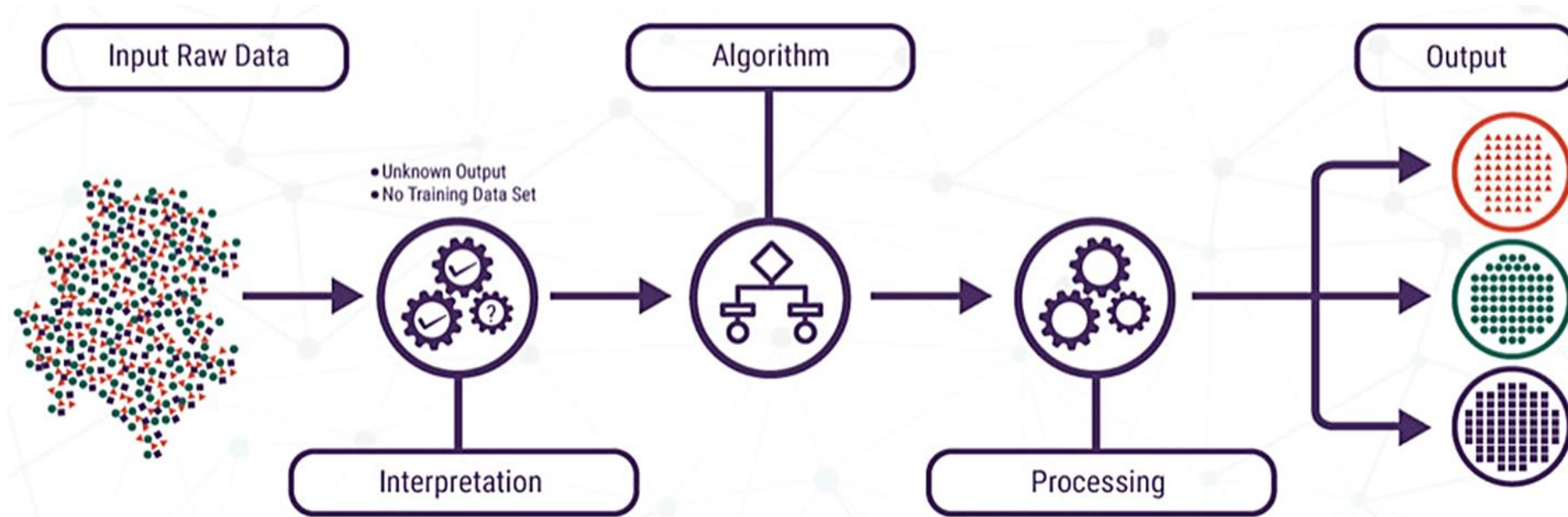
Spam Detection: Classifying emails as spam or not spam.

Image Classification: Identifying objects within images, such as detecting faces.

Predictive Maintenance: Predicting when a machine is likely to fail based on historical data.

Unsupervised Learning

- Unsupervised learning deals with unlabelled data.
- The algorithm identifies patterns without predefined answers.
- The goal is to discover structures, groupings, or relationships within the data.



Source: [van Loon, R. \(2024\)](#)

Unsupervised Learning

Data Requirement: Works with unlabelled data.

Model Types: Clustering, Dimensionality Reduction, and Anomaly Detection.

Learning Process: The model explores the data to identify patterns or groupings without explicit guidance on what to look for.

Evaluation: Evaluation is less straightforward than in supervised learning, often relying on metrics like silhouette scores or visual inspection of clusters.

Applications:

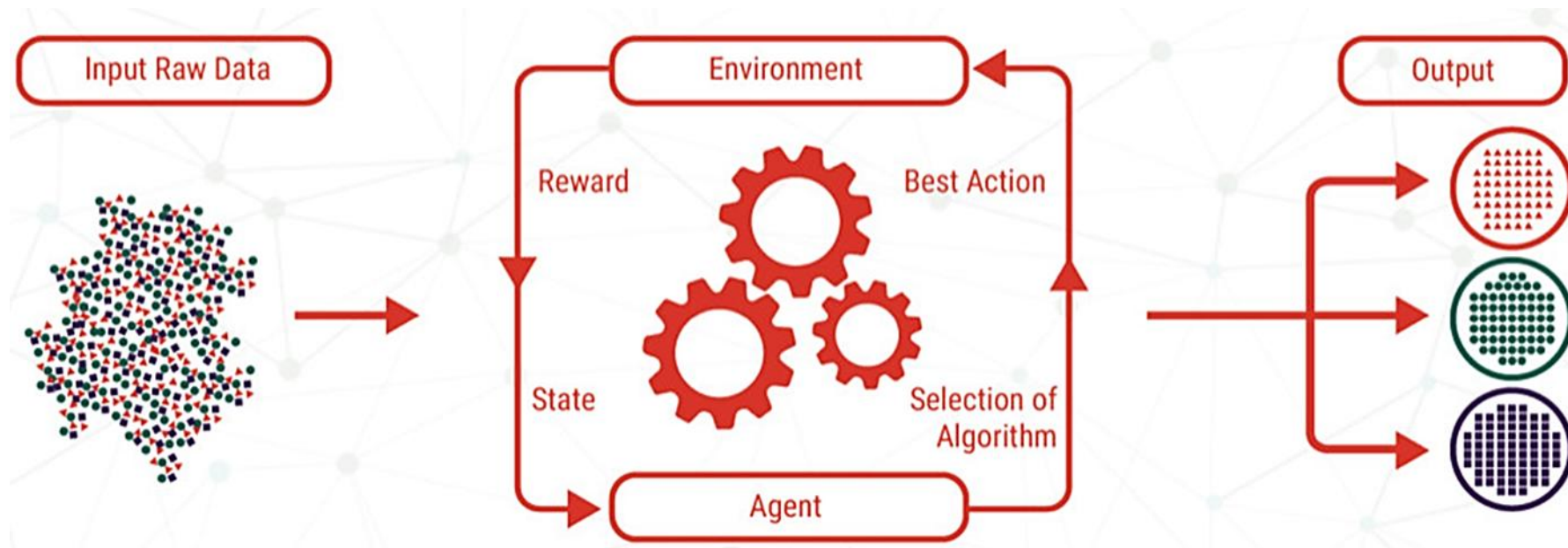
Customer Segmentation: Grouping customers based on purchasing behaviour.

Market Basket Analysis: Identifying items frequently purchased together.

Anomaly Detection: Detecting outliers in data, such as fraudulent transactions.

Reinforcement Learning

- An intelligent agent learns to make decisions by performing actions in an environment and receiving rewards or penalties in return.
- The goal of the agent is not to find hidden patterns in data (like in Unsupervised Learning) or to predict a known label (like in Supervised Learning). Instead, the agent learns through trial and error to discover the optimal sequence of actions that will maximize its cumulative long-term reward.



Source: [van Loon, R. \(2024\)](#)

Reinforcement Learning

Data Requirement: The agent learns through trial and error, interacting with the environment.

Model Types: Q-learning, SARSA, deep Q-networks (DQN) and policy gradient methods.

Learning Process: The agent balances exploration (trying new actions) and exploitation (choosing the best-known actions) to learn the optimal policy.

Evaluation: Performance is evaluated based on the cumulative reward obtained by the agent.

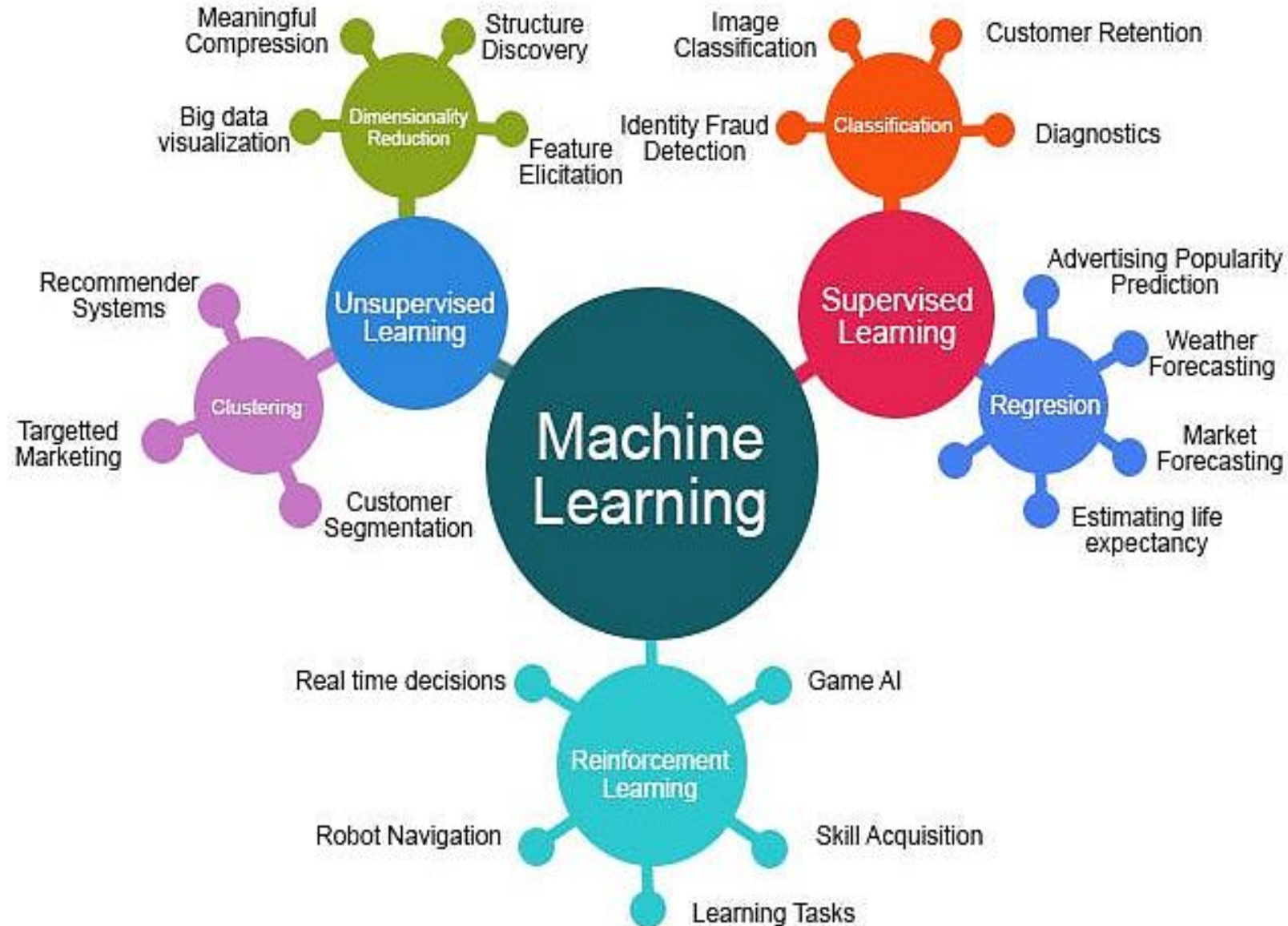
Applications:

Game Playing: Agents learning to play games like chess, Go, or video games at superhuman levels.

Robotics: Teaching robots to perform tasks like grasping objects or walking.

Autonomous Vehicles: Developing self-driving cars that can make decisions in real-time based on sensor data.

Overview



Source: www.educba.com

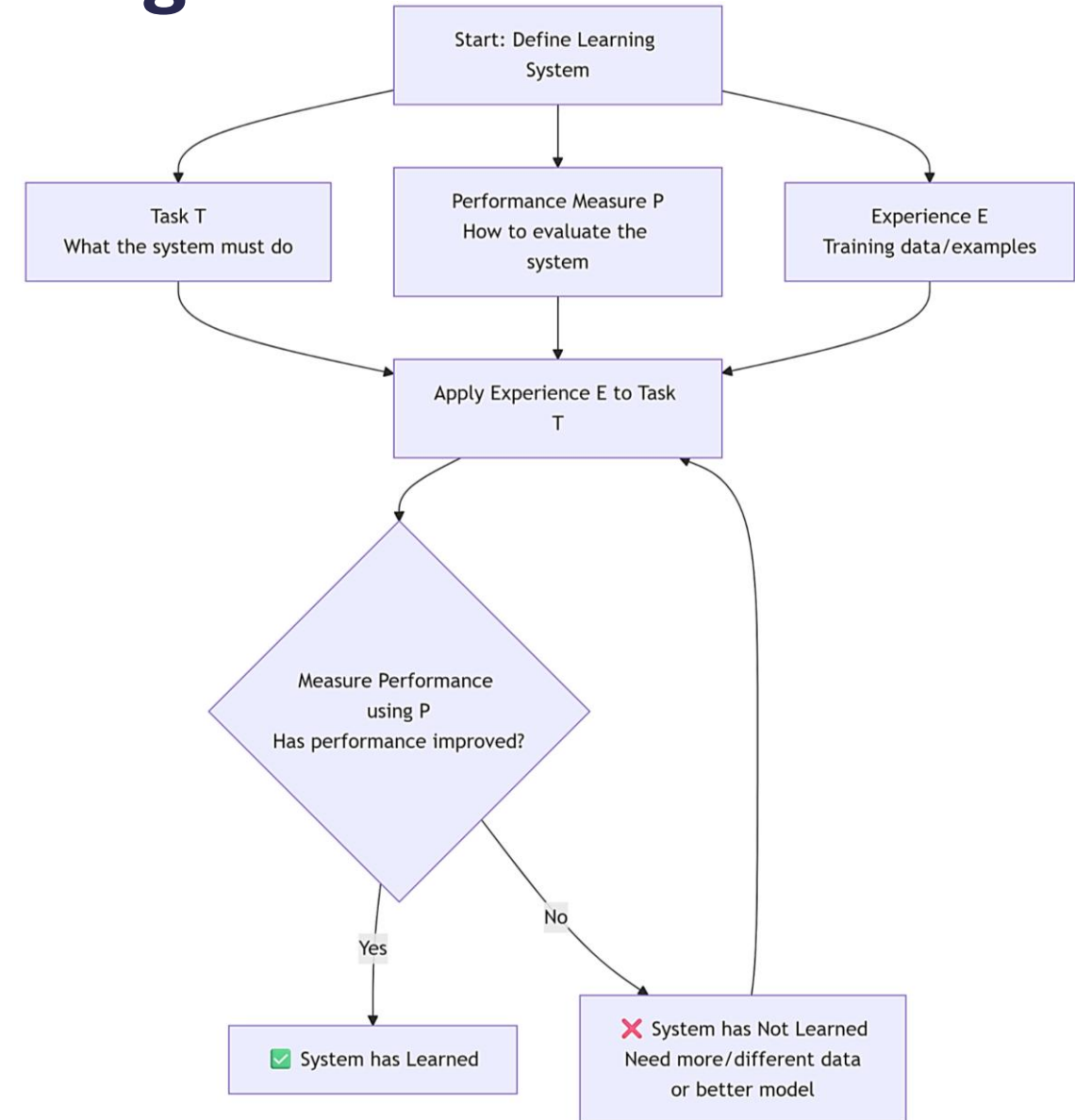
A Comparison

Aspect	Supervised Learning	Unsupervised Learning	Reinforcement Learning
Data Type	Labelled data (input–output pairs)	Unlabelled data (only inputs)	Sequential data with states, actions, and rewards
Purpose	Learn a mapping from inputs to known outputs for prediction or classification	Discover hidden structures, patterns, or relationships in data	Learn an optimal policy to maximize cumulative rewards
Output Type	Discrete labels or continuous values	Clusters, groups, or latent representations	Sequence of optimal actions or policies
Complexity	Moderate to high depending on model and data size	Moderate; interpretation can be challenging	Very high; requires exploration, simulation, and long training times
Training Data Requirement	Requires large, accurately labeled datasets	Requires large volumes of raw, unlabeled data	Requires many interactions with an environment (real or simulated)
Human Intervention	Needed for labeling and supervision	Minimal; mostly during interpretation	Needed for environment design and reward shaping
Learning Goal	Minimize prediction error	Find meaningful data structure	Maximize cumulative reward through exploration and exploitation
Example Algorithms	Linear Regression, Logistic Regression, Decision Trees, SVM, Neural Networks	K-Means Clustering, Hierarchical Clustering, PCA, Autoencoders, DBSCAN	Q-Learning, Deep Q-Networks (DQN), SARSA, Policy Gradients, Actor-Critic
Common Use Cases	Image classification, spam detection, medical diagnosis, sales forecasting	Customer segmentation, market basket analysis, anomaly detection, data compression	Game playing (e.g., AlphaGo), robotics control, autonomous driving, recommendation systems

Formal Definition of Machine Learning

A computer system is said to learn from some *experience E* with respect to some *task T* and *performance measure P* , if its performance at task T , as measured by P , improves with experience E .

Mitchell (1997)



Experience (E)

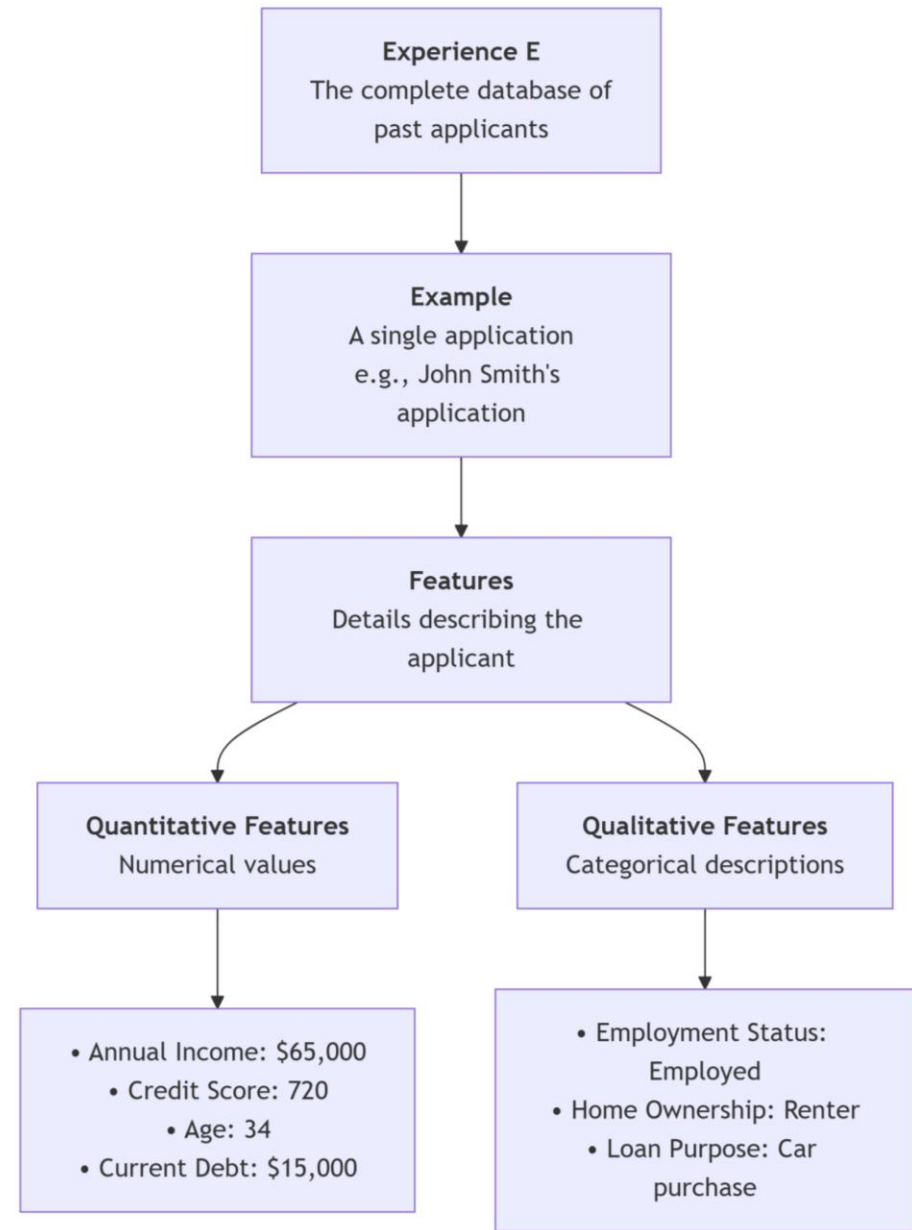
Experience: A collection of examples related to a specific task.

Example: A single item or event in the dataset. Each example is described by its features.

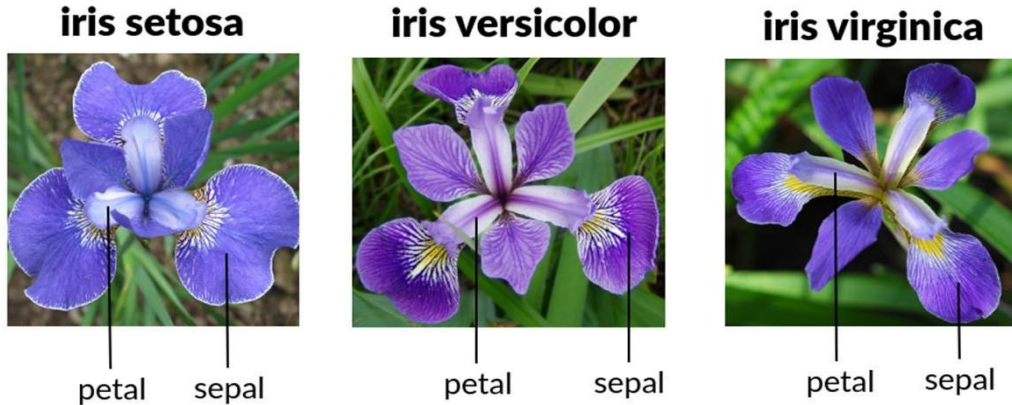
Features: Are the specific details or characteristics that describe the example, can be quantitative or qualitative.

Examples of features:

- Age, Purchase History, Location (**features**) of a customer (**example**)
- Employment status, Housing status, annual income (**features**) of a credit application (**example**)
- Temperature, Humidity, Chance of Rain (**features**) of a day (**example**)



Experience (E)



https://s3.amazonaws.com/assets.datacamp.com/blog_assets/iris-machinelearning.png

Article in which the IRIS database was first introduced:

Fisher, R.A. (1936) "The use of multiple measurements in taxonomic problems" Annual Eugenics, 7, Part II, 179-188

The Iris dataset is perhaps the most well-known database in the field.

Dataset Structure:

150 examples perfectly balanced across 3 iris species:

- Setosa
- Versicolor
- Virginica

Feature Description:

Each example is characterized by 4 botanical measurements:

- Sepal length (cm)
- Sepal width (cm)
- Petal length (cm)
- Petal width (cm)

Task (T)

Refers to the specific activity the system is designed to perform.

Defines what the machine learning model is trying to accomplish.

Examples of ML tasks:

- **Classification:** Assigning labels to inputs (e.g., spam detection).
- **Regression:** Predicting continuous values (e.g., house price prediction).
- **Clustering:** Grouping similar data points (e.g., customer segmentation).
- **Recommendation:** Suggesting items to users (e.g., movie recommendations).
- **Anomaly Detection:** Identifying unusual data points (e.g., fraud detection).

The choice of task determines the learning approach and performance evaluation.

Performance Measure (P)

- A quantitative measure of model effectiveness.
- Used to evaluate how well a machine learning algorithm performs a given task.
- Depends on the task type (e.g., classification, regression, clustering).
- Typically calculated on unseen test data to assess generalization ability.
- Helps compare models, tune hyperparameters, and guide improvement.

Performance Measure (P)

Regression Performance Measures

Assess how close predicted values are to actual continuous values.

Key Metrics:

Mean Squared Error (MSE): Average of squared prediction errors.

Root Mean Squared Error (RMSE): Square root of MSE, interpretable in original units.

Mean Absolute Error (MAE): Average of absolute prediction errors.

R^2 Score (Coefficient of Determination): Proportion of variance explained by the model.

Adjusted R^2 : Adjusted for number of predictors, avoids overfitting.

Goal: Minimize prediction errors and maximize explained variance.

Generalization in Machine Learning

Ability of a model to perform well on **new, unseen data**.

Indicates the model has learned **underlying patterns**, not just memorized training data.

Prevents overfitting and avoids high performance on training data but poor results on test data.

Evaluation: Tested on a **separate test dataset** or using **cross-validation**

Goal: Achieve a balance between **training accuracy** and **test accuracy** for reliable real-world performance.

Underfitting and Overfitting

Underfitting:

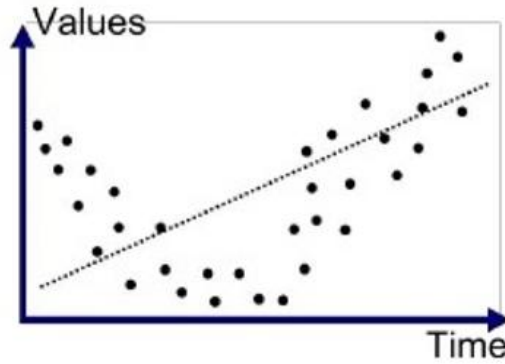
- Model is **too simple** to capture underlying patterns in the data.
- Cannot achieve a **low error on the training set**.
- Performs **poorly on both training and test data**.
- Causes: insufficient features, overly simple model, too little training.
- Solution: increase model complexity, add relevant features, train longer.

Overfitting:

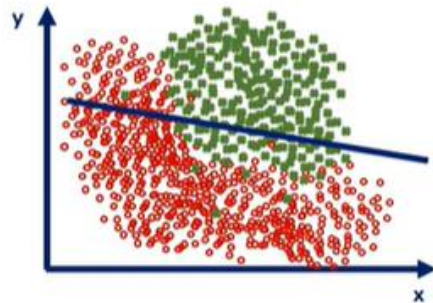
- Model is **too complex** and memorizes training data instead of learning patterns.
- Low training error, **high test error** (large gap between training and test errors).
- Performs **very well on training data** but poorly on unseen test data.
- Causes: overly complex model, too many features, insufficient training data.
- Solution: regularization, simplify model, use more training data, cross-validation.

Goal: Find the right model complexity to **minimize both training and test errors**, achieving good **generalization**.

Underfitting and Overfitting

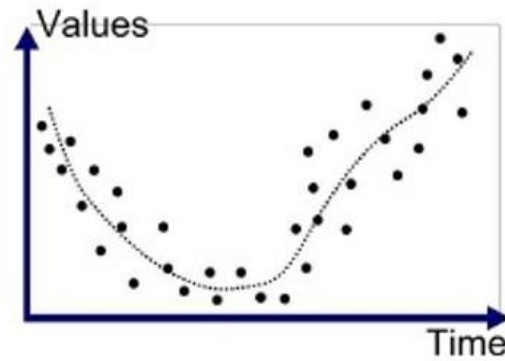


Underfitted

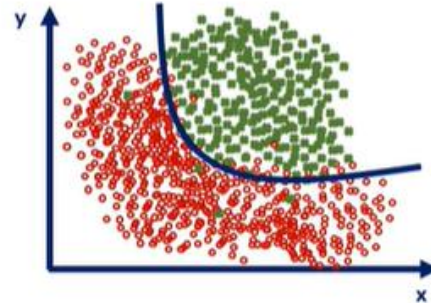


Doesn't capture any logic

- High loss
- Low accuracy

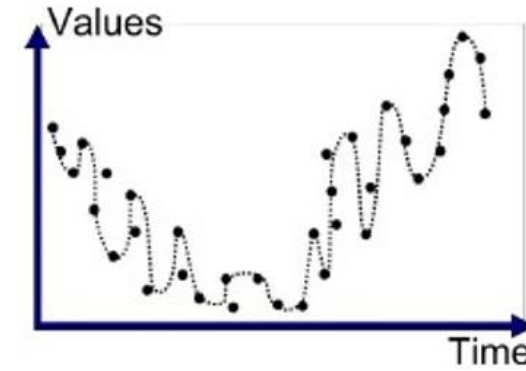


Good Fit/Robust

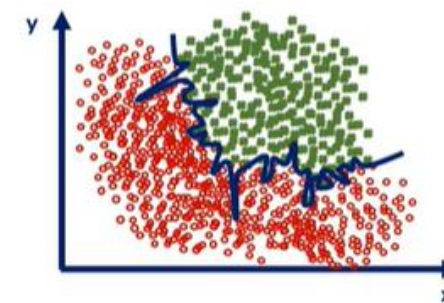


Captures the underlying logic of the dataset

- Low loss
- High accuracy



Overfitted



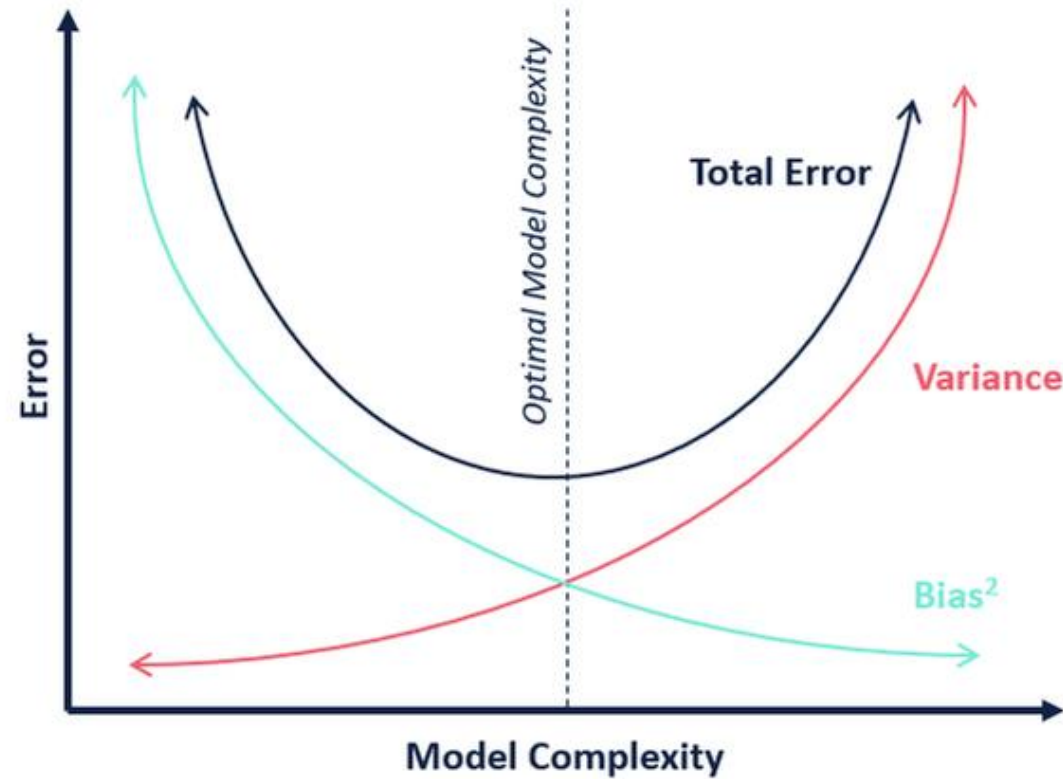
Captures all the noise, thus "missed the point"

- Low loss
- Low accuracy

Source: www.medium.com

Goal: Find the right model complexity to **minimize both training and test errors**, achieving good generalization.

Underfitting and Overfitting



Models with **underfitting** usually have **high bias and low variance**. It happens when we have very less amount of data to build an accurate model

underfitting (high bias) optimal balance overfitting (high variance)

Models with **overfitting** have **low bias and high variance**. It happens when the model captures the noise along with the underlying pattern in data

Model Capacity

Ability of a model to fit a wide variety of functions.

Controls the likelihood of **underfitting** or **overfitting**.

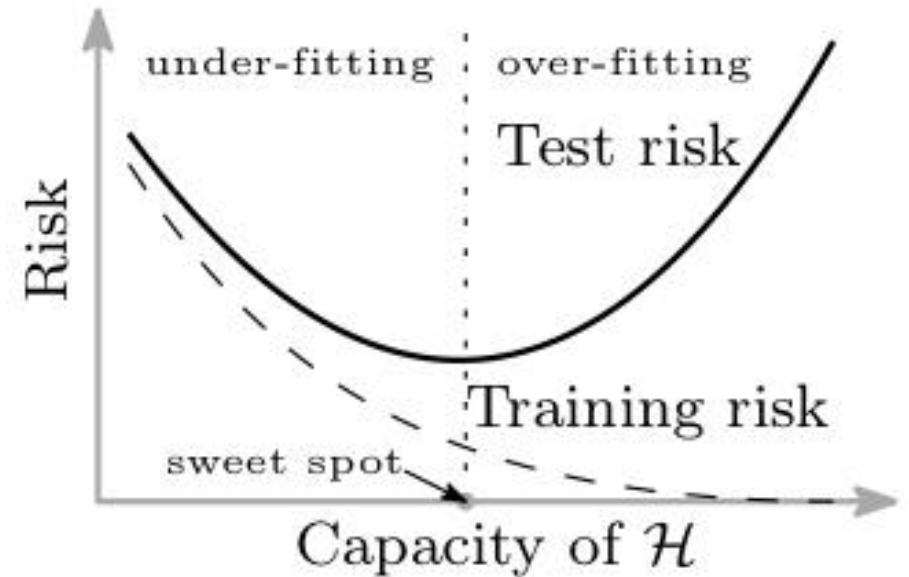
Low-capacity model:

- Too simple → cannot fit training data well → **underfitting**.
- High training error and high-test error.

High-capacity model:

- Too complex → memorizes training data → **overfitting**.
- Low training error but high-test error.

Goal: Choose **optimal model capacity** to achieve **good generalization**.



Source: [Belkin et al., 2019](#)

A Case Study

From Data Overload to Smart Predictions

How supermarkets can use historical data to forecast future sales



The Goal

Learn from the Past
to Predict the Future

A supermarket has
millions of transactions.

Problem

Raw data is too big and
complex to use directly.

Solution

Use Machine Learning to find
patterns and predict future
sales.



The Challenge

Supermarkets store millions of transactions,
but raw data is impossible to manage
directly.

Objective: Use Machine Learning to find
patterns and predict future product sales.



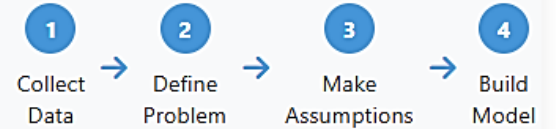
Data We Use

To predict sales for a future day, we
summarize past days with:

- **Temporal Data:** Previous sales, Day of week
- **Environmental:** Temperature, Weather conditions
- **Economic Factors:** Price, Price changes, Unemployment rate



The ML Process



Key assumptions: Customer behaviors are similar
year-to-year, purchase decisions are independent.

A Case Study



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Sales Prediction Method

Using Nearest Neighbor Classifier to Forecast Product Sales



Data Representation

We summarize past experiences using
these key attributes:

Attribute	Description	Type
sale_f-1	Previous day's sales	Numerical
week_day	Day of the week	Categorical
temperature	Temperature in °C	Numerical
general_conditions	Weather conditions	Categorical
price	Product price in €	Numerical
price_change	Price change	Categorical
unemployment_rate	Unemployment rate	Numerical
Euribor	Euro rate	Numerical

$description(f) = (sale_f-1, week_day, temperature, general_conditions, price, price_change, unemployment_rate, Euribor)$

Target Variable: $sale(p) = quantity_sold$



Training Set

Our model learns from historical data
organized as:

Training Set:
 $\{[description(f_i), sale(p_i)]\}$

Where:

- **$description(f_i)$** = Attribute vector for date f_i
- **$sale(p_i)$** = Actual quantity sold
- Each pair = one historical experience
- Model finds patterns in these examples



Nearest Neighbor Prediction

Simple method for sales forecasting:



Method: Find relationship between inputs
and outputs

Given future date f and $description(f)$:

1. Find closest $description(f_i)$ in training set
2. Use sales from that day as prediction

$estimation: sale(p) \leftarrow sale(p_i)$

A Case Study

Building & Improving Our Sales Predictor

From simple predictions to refined models



The Goal

Learn from the Past
to Predict the Future

A supermarket has
millions of transactions.

Problem

Raw data is too big and
complex to use directly.

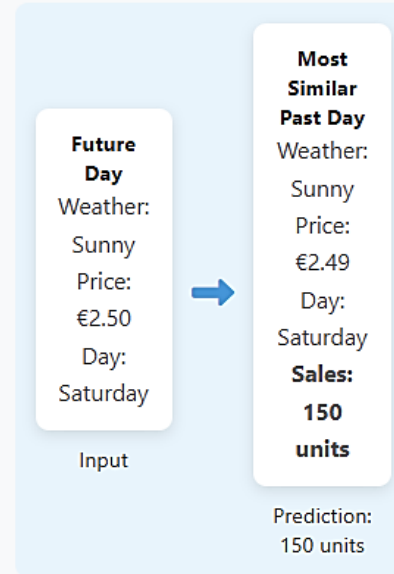
Solution

Use Machine Learning to find
patterns and predict future
sales.



The Prediction Method

"Find the Most Similar Day"
(Nearest Neighbor Algorithm)



We search through historical data to
find the day that looks most similar to
our target day, then use its sales as
our prediction.



Evaluation & Testing

We measure prediction quality by:

- Splitting data into **Training Set** (to build the model) and **Test Set** (to check accuracy)
- Calculating error metrics like **Mean Squared Error (MSE)**
- Comparing predicted vs. actual sales



Refinement & Improvement

We can improve predictions by
enhancing:



Better Data
Add holidays,
promotions,
product relationships



Smarter Models
Use more complex
algorithms



Improved Assumptions
Consider product
bundles (beer + chips)

THANK YOU

Any Questions?

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